

# Problem Session 1

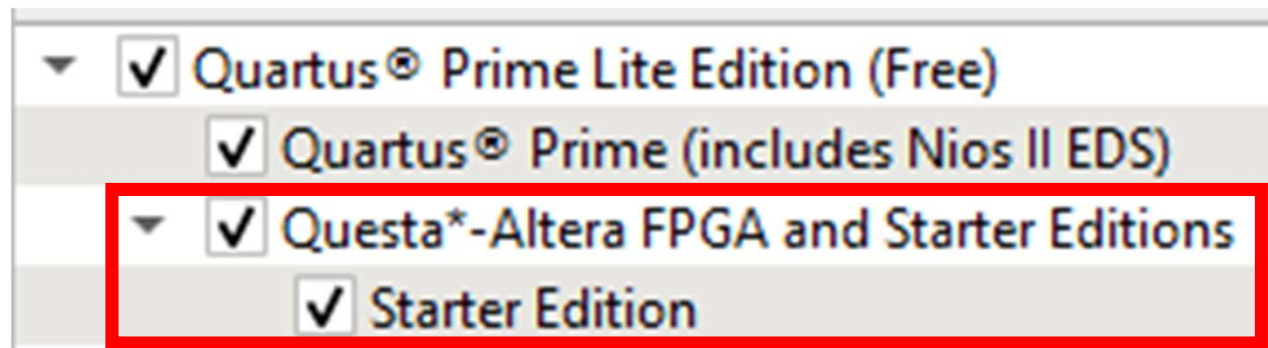
Gebze Technical University  
Computer Engineering Department  
CSE331 – Computer Organization  
Enes Abdülhalik

# Contents

- Quartus Installation
- Questa License Installation
- Verilog Modules
- Verilog Coding Models:
  - Structural
  - Dataflow
  - Behavioral

# Quartus Installation

- Note: Questa may not run on Linux. Windows is more preferred.
- We will be using Quartus Prime Lite Edition 24.1.
- Do not forget to install Questa starter edition alongside it.
- Install it normally, it won't require any licensing.



# Questa License Installation

- Quartus 24.1 has an option for automatic license setup for Questa.
- On older versions, you need to create a license using Intel Self Service Licensing Center.

# Verilog Module

```
module sample (  
    a, b, c, d  
);  
  
input a, b, c;  
output d;  
  
//Logic  
  
endmodule
```

```
module sample (  
    input a, b, c,  
    output d  
);  
  
//Logic  
  
endmodule
```

# Structural Model

- Allows the introduction of hierarchy in a design.
- Modules can be instantiated inside other modules.

```
'and and1(a_and_b, a, b);
```

# Dataflow Model

- C-like syntax for logic operations.
- Includes most of the basic logic operations.
  - AND, OR, XOR, 2 to 1 MUX, etc.

```
wire a_and_b;  
assign a_and_b = a & b;
```

# Behavioral Model

- Allows using:
  - Blocking and non-blocking statements.
  - Timing delays.
  - If-else and case statements.
  - Clock signals for sequential circuits.

```
always @ (*Sensitivity List*/) begin
// Statements
end

initial begin
// Statements
end
```



Thanks for listening